

WORK BREAKDOWN STRUCTURE (WBS)

Second-Hand Product Escrow Marketplace #39

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5TH SEM 'H'

1. Project

Second-Hand Product Escrow Marketplace

2. WBS Objective

The Work Breakdown Structure decomposes the project into manageable work packages covering requirements, escrow processing, inspection, buyer verification, dispute handling, security, testing, and project documentation.

3. Work Breakdown

WBS ID	Work Package / Task	Description
1.0	Requirements Engineering	Define and organize the system requirements.
1.1	Identify stakeholders	Identify Buyer, Seller, Inspection Agent, Administrator, and payment/escrow service.
1.2	Define functional requirements	Document FR-001 through FR-005.
1.3	Define non-functional requirements	Document NFR-001 and NFR-002.
1.4	Define acceptance criteria	Specify measurable conditions for each requirement.
2.0	Escrow Management	Implement the core escrow transaction lifecycle.
2.1	Fund order	Receive/record buyer payment and create the escrow state.
2.2	Lock escrow payment	Prevent release before the required verification step.
2.3	Track escrow status	Maintain order states such as Under Inspection and release/frozen states.
2.4	Release/freeze funds	Release funds under valid conditions and freeze them when a dispute is raised.
3.0	Product Inspection	Support inspection and evidence submission.
3.1	Assign/perform inspection	Inspection agent performs the product inspection.
3.2	Record pass/fail result	Store the inspection outcome.

3.3	Upload photographic evidence	Attach inspection photographs to the report.
4.0	Buyer Verification	Allow the buyer to make the required decision.
4.1	Display inspection report	Show inspection result and evidence to the buyer.
4.2	Approve item	Record buyer approval.
4.3	Reject item	Record buyer rejection and trigger the applicable flow.
4.4	Handle 48-hour timeout	Apply the automatic-release rule after a successful inspection when the buyer does not respond.
5.0	Dispute Management	Handle transaction disputes safely.
5.1	Raise dispute	Allow buyer to initiate a dispute.
5.2	Freeze escrow	Prevent fund release/withdrawals while a dispute is active.
5.3	Resolve dispute	Administrator manages the resolution process.
6.0	Security	Protect sensitive marketplace and escrow information.
6.1	Encrypt sensitive data	Protect payment and escrow-related information.
6.2	Secure transactions	Apply appropriate access and transaction controls.
7.0	Testing and Validation	Verify that the system satisfies requirements.
7.1	Functional testing	Test each functional requirement.
7.2	Exception-flow testing	Test payment, inspection, rejection, timeout, and dispute exceptions.
7.3	Security testing	Verify protection of sensitive data and escrow operations.
7.4	Acceptance validation	Verify acceptance criteria for all requirements.
8.0	Documentation and Project Management	Maintain project deliverables and development records.
8.1	SRS	Prepare and maintain the Software Requirements Specification.
8.2	UML and architecture documentation	Maintain use-case and architecture/design artifacts.
8.3	Jira/Scrum/Kanban tracking	Track stories, tasks, subtasks, and project progress.
8.4	GitHub repository	Maintain source code, project files, and version history.
8.5	GitHub Copilot evidence	Capture screenshots demonstrating relevant Copilot usage.

4. Traceability to Project Requirements

The WBS is aligned with the project's defined requirements: FR-001 to FR-005 and NFR-001 to NFR-002. The escrow, inspection, buyer verification, release, dispute, security, and testing work packages provide the major implementation and validation areas needed to satisfy those requirements.